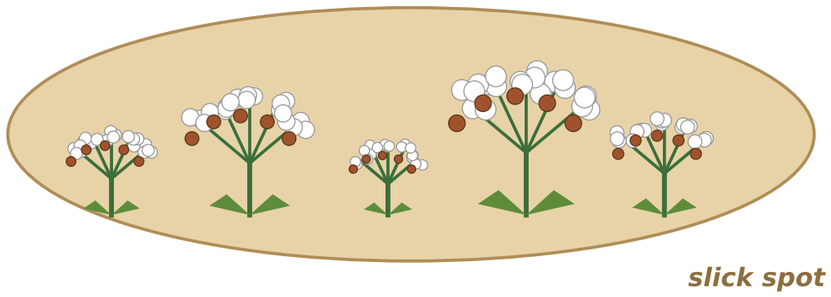
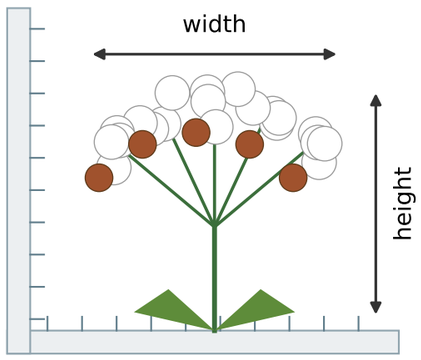
1. Plant phenotyping

every flowering plant



slick spot



plant size

seed yield

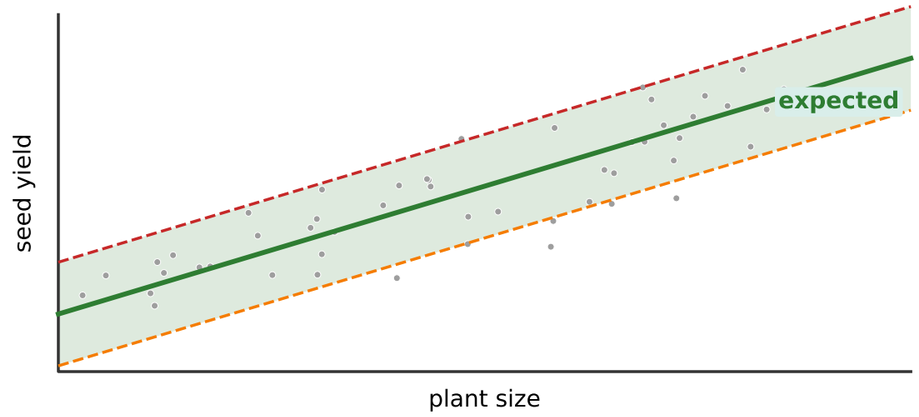
fast & low cost

2025 2026

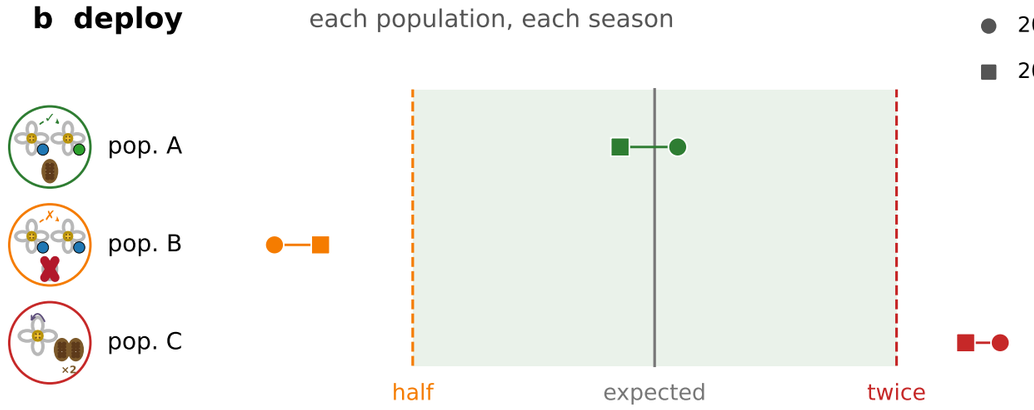
plant size + seed yield, every plant, two seasons

2. Predicting reproductive status

a build all plants → seed yield expected for size

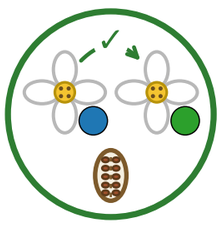


b deploy each population, each season



build on all plants → deploy by population

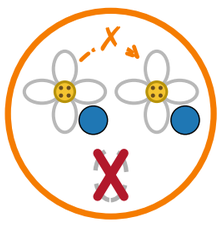
3. Sort populations for restoration



reproductively intact

→ source for seed bulking

top priority



mate-limited

→ genetic rescue

flaw: reduced fecundity

rescue



over-producing

→ excluded as source

flaw: inbreeding depression

avoid

hypotheses → tested with genetics (panel 4)

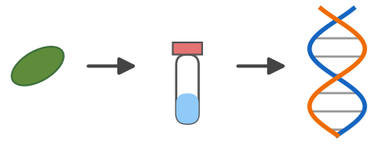
4. Test with genetics

Why test?

seed yield can also be driven by

resource or pollinator availability

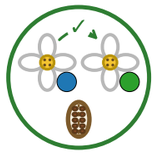
our research: pollinators are not limiting → genetics



SRK gene:

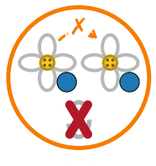
the S-types of every plant

what genetics should show



reproductively intact

many S-types, many compatible mates



mate-limited

few S-types, few compatible mates



over-producing

seeds carry only the mother's S-types (selfing)